

Assignment 1: Network Latencies, Ping, and Traceroute

Your Name
Your Purdue ID

CS 422 – Fall 2026

Repository

The complete source code, input data, and instructions needed to reproduce the experiments and plots are available at:

<https://github.com/YOUR-USERNAME/YOUR-REPOSITORY>

Reproduction command:

```
./run_all.sh
```

1 Ping Test and Round-Trip Time

1.1 Method

We tested every destination listed in the provided IP-address file, along with our own public IP address. [Describe the rest of the procedure here.]

The code used to collect and process the ping measurements is available here: [ping collection](#) and [RTT calculation](#).

1.2 Results

Table 1: Example format for selected ping results.

Destination IP	Distance (km)	Min RTT (ms)	Avg RTT (ms)	Max RTT (ms)
Add results here				

1.3 Distance versus RTT

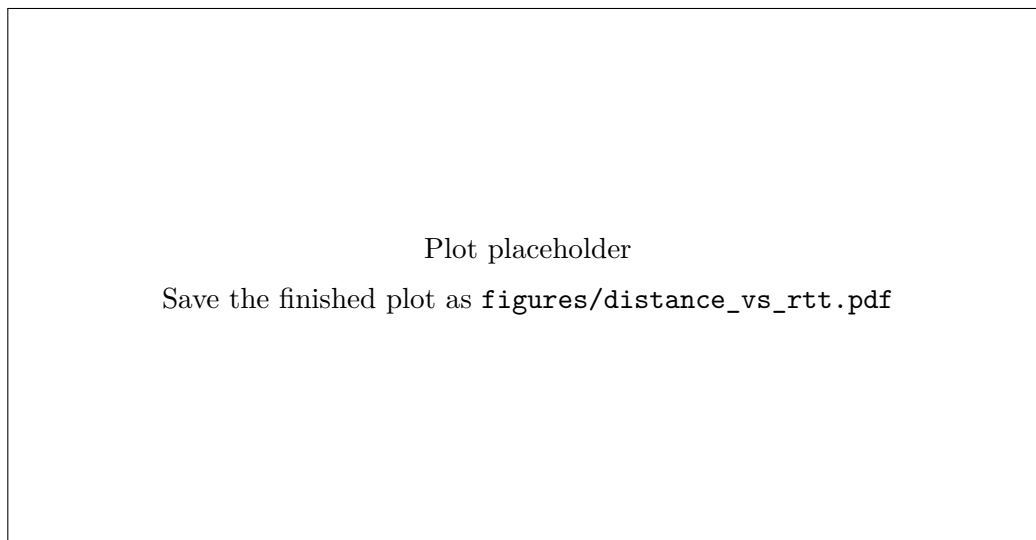


Figure 1: Geographic distance from West Lafayette, Indiana, versus round-trip time. Each point represents one destination IP address.

The plotting code is available here: [distance-versus-RTT plot](#).

1.4 Discussion

[Explain how distance relates to RTT. Discuss propagation delay, routing, congestion, queueing, processing delay, and any important outliers. Explain what the minimum and maximum RTT values convey.]

2 Latency Breakdown

2.1 Method

Table 2: Five destinations selected for traceroute analysis.

Destination IP	Site/Country	Selection notes
Add five destinations here		

[Describe how traceroute was run, how RTT was obtained for each intermediate hop, and how non-responsive hops marked with * were filtered or recorded.]

The traceroute collection code is available here: [traceroute collection and parsing](#).

2.2 Per-Hop Latency Breakdown

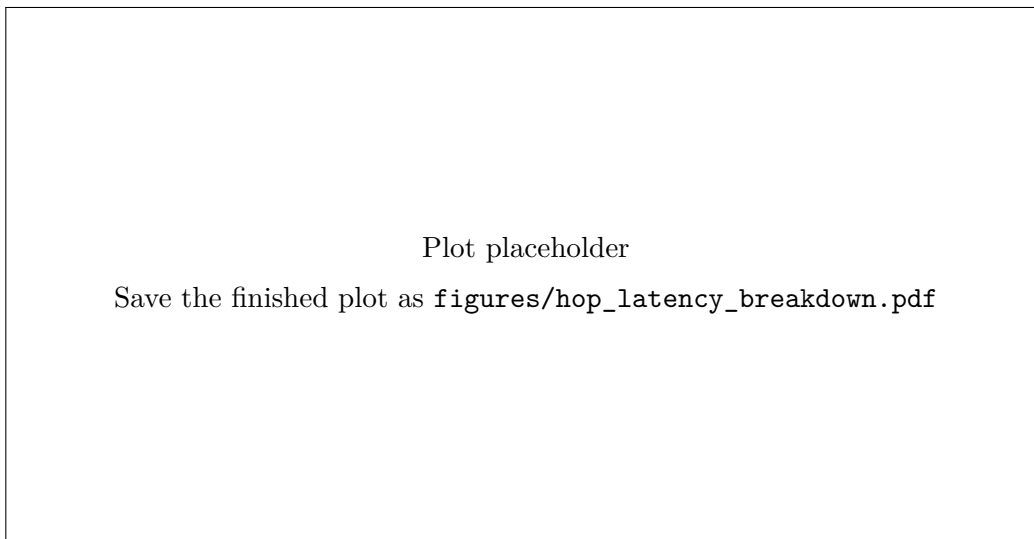


Figure 2: Stacked latency breakdown for the responsive hops along the paths to the five selected destinations.

The plotting code is available here: [stacked per-hop latency chart](#).

2.3 Hop Count versus RTT

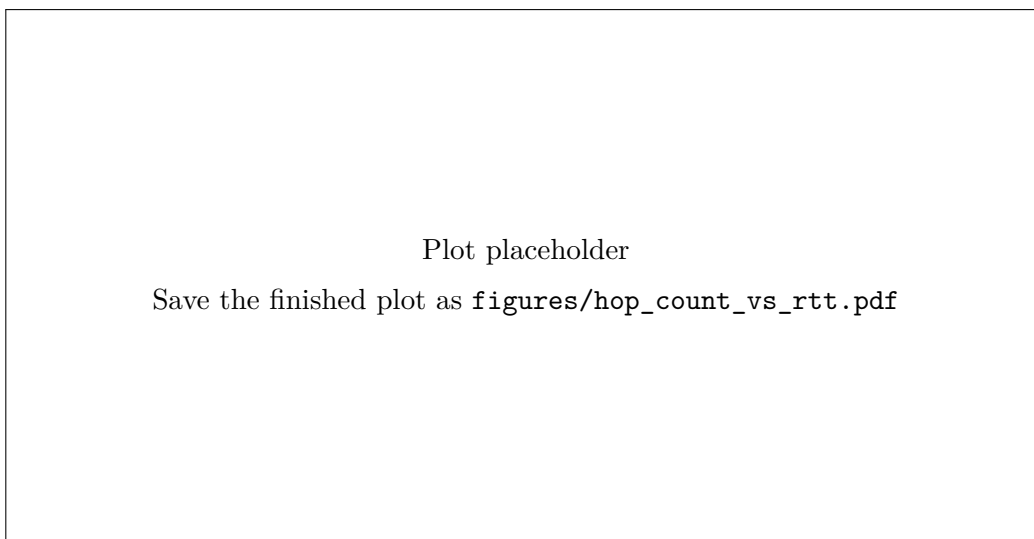


Figure 3: Hop count versus destination RTT. Each point represents one of the five selected destination IP addresses.

The plotting code is available here: [hop-count-versus-RTT plot](#).

2.4 Discussion

[Explain the major observations from both plots and discuss how hop count relates to RTT. Mention relevant effects such as geographic path length, link speed, routing, congestion, and routers that do not answer ICMP probes.]

3 Conclusion

[Briefly summarize the main relationship observed among geographic distance, hop count, and RTT.]